Hematological Evaluation of the Toxicity of *Icacina Trichantha* Ethanolic Root Extract in Rats

[*1]Eseine-Aloja C.E., [2]Ugbeni O.C., [3]Ujaddughe M.O., [4]Shelu J.O.

[1]Department of Medical Biochemistry, Faculty of Basic Medical Sciences, Ambrose Alli University, Ekpoma, Edo State, Nigeria.
[2]Department of Biochemistry, Faculty of Life Sciences, University of Benin, Benin City, Nigeria.
[3] Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa.
[4]Department of Human Physiology, Ambrose Alli University, Ekpoma, Faculty of Basic Medical Sciences, Edo State, Nigeria

ABSTRACT

Icacina trichantha is widely used in traditional medicine; however, its haematological safety profile remains inadequately characterized. This study evaluated the haematological effects of sub-acute oral administration of *Icacina trichantha* root ethanolic extract (ITREE) in rats. Twelve adult male Wistar rats were allocated into four groups; control, 200, 500, and 1000 mg/kg ITREE treated (orally) for 28 days. Thereafter, blood samples were collected for haematological analysis using an automated haematology analyzer. Data were analysed using ANOVA followed by the Least Significant Difference post-hoc test, with statistical significance set at $p < 0.05$. Administration of ITREE did not produce significant changes in red blood cell count, haemoglobin concentration, total white blood cell count, or platelet count. However, significant dose-dependent reductions were observed in haematocrit (43.30 - 44.23% vs. 47.33%) and mean corpuscular volume (56.27 - 59.97 fL vs. 62.36 fL), while mean corpuscular haemoglobin concentration increased significantly at the highest dose (37.08 g/dL vs. 33.83 g/dL). These findings indicate that sub-acute administration of ITREE at lower doses does not adversely affect major haematological parameters, suggesting a favourable haematological safety profile under the conditions of this study. Nevertheless, the dose-dependent alterations in erythrocyte indices observed at 500 and 1000 mg/kg may indicate early microcytic changes associated with higher levels of exposure. Further long-term and mechanistic studies are therefore recommended to elucidate the underlying mechanisms and establish the chronic haematological safety of the extract.

Keywords: Erythropoiesis, Hematology, *Icacina trichantha*, Toxicity, Wistar rats.

***Corresponding author.** Tel. +2348037950022. **E-mail address:** cee-aloja@aauekpoma.edu.ng

1. INTRODUCTION

Icacina trichantha Oliv. (family Icacinaceae) is an understorey, drought-resistant shrub indigenous to the tropical forests of West and Central Africa [1,2]. In Nigeria, it is commonly known by different indigenous names, including *Urumbia* or *Eriagbo* among the Igbo, reflecting its strong emetic properties; *Gbegbe* (meaning "to cleanse") among the Yoruba; and *Ebosan* (meaning "to energize") among the Esan people of Edo State. The plant thrives in forest ecosystems but has also been reported as a weed in rice paddies. It grows to approximately two meters in height and possesses large, fleshy, yam-like underground tubers that may weigh several kilograms [3]. The fruit is a drupe containing an edible soft pulp, while the starchy tubers have traditionally served as an important famine food and are consumed either fresh or processed into flour for the preparation of soups, pastes, and porridges [4]. Nutritional analyses have shown that the tubers are rich in carbohydrates and contain appreciable amounts of proteins, lipids, and essential minerals. However, they also contain anti-nutritional constituents such as hydrogen cyanide, oxalates, tannins, phytates,

and alkaloids, which may influence their nutritional and toxicological properties [4,5].

Beyond its nutritional value, *I. trichantha* occupies an important place in African traditional medicine. In southwestern Nigeria, various parts of the plant are used as first-aid remedies for food poisoning [6]. The leaves and seeds are traditionally employed in the management of hypertension and asthma [7], whereas the tubers are used in the treatment of malaria, constipation, poisoning, toothache, and rheumatic conditions, and are also administered as emetics and abortifacients [1,8]. In addition, preparations made from the tubers and leaves have been reported to be useful in the management of mumps and soft tissue swellings, while some communities attribute aphrodisiac properties to the plant [3,9,10].

The extensive ethnomedicinal use of *I. trichantha* has stimulated considerable scientific interest in its pharmacological activities. Earlier investigations demonstrated its antidiarrhoeal activity [11], central nervous system depressant effects through prolongation of pentobarbitone-induced sleeping time, local anaesthetic properties, and anticonvulsant activity [1,12,13]. More recent studies have reported a broad spectrum of biological activities, including hepatoprotective, antigenotoxic, cytotoxic, antidiabetic, and insecticidal effects [14–18]. These findings suggest that *I. trichantha* contains biologically active phytochemicals with potential therapeutic applications.

Despite these promising pharmacological properties, the widespread use of medicinal plants raises important concerns regarding their safety. Herbal preparations may contain bioactive constituents capable of producing adverse effects, particularly when consumed in high doses or over prolonged periods [19,20]. Consequently, comprehensive toxicological evaluation is essential to establish their safety profile, determine appropriate dosage ranges, and support their rational therapeutic use, in line with the World Health Organization (WHO) recommendations for the safe use of herbal medicines [21,22].

Experimental animal models remain indispensable for assessing the toxicological effects of medicinal plants because they permit controlled evaluation of exposure-related physiological and pathological changes [20,23,24]. Among the various indices of systemic toxicity, hematological parameters are particularly valuable because they are highly regulated biological indicators that respond rapidly to physiological disturbances and toxic insults. Alterations in hematological indices may therefore provide early evidence of adverse effects on hematopoiesis, immune function, or overall systemic health [25].

Although several studies have documented the pharmacological properties of *I. trichantha*, information regarding its hematological safety remains limited. The present study was therefore undertaken to evaluate the hematological effects of sub-acute oral administration of *Icacina trichantha* root ethanolic extract (ITREE) in Wistar rats. The findings are expected to contribute to the growing body of evidence on the safety profile of this medicinal plant and provide useful information for its potential therapeutic application.

2. MATERIALS AND METHODS

2.1. Experimental animals: This study employed an *in vivo* experimental design using adult Wistar rats. Twelve healthy adult Wistar rats, each weighing approximately 150 g, were obtained from the Animal House of the Department of Biochemistry, University of Benin, Benin City, Nigeria. The animals were acclimatized for 14 days under standard laboratory conditions, maintained on a 12-hour light/12-hour dark cycle, and provided with standard laboratory rat chow and clean drinking water *ad libitum* throughout the experimental period.

2.2. Ethical approval: Ethical approval for this study was obtained from the Faculty of Life Sciences Research Ethics Committee (FLSREC), University of Benin, Benin City, Nigeria (Approval No. FLSRE-2023-016). All experimental procedures involving animals were conducted in accordance with established institutional guidelines for the care and use of laboratory animals.

2.3. Plant material: Fresh roots of *Icacina trichantha* were collected from Idunwele-Ewu, Esan Central Local Government Area, Edo State, Nigeria. Botanical authentication was carried out at the Herbarium of the Department of Botany, Ambrose Alli University, Ekpoma, Edo State, Nigeria, where a voucher specimen (No. 1046) was deposited for future reference. All analytical-grade reagents and chemicals used in this study were obtained from Pyrex LG Scientific Supply Company, Benin City, Nigeria (RC-1911615).

2.4. Preparation of *Icacina trichantha* root ethanolic extract: The freshly collected roots were thoroughly washed with clean water to remove adhering soil and debris, air-dried at room temperature, and pulverized into a fine powder using a mechanical grinder. Twenty kilograms (20 kg) of the powdered root material were macerated in 40% ethanol at a solvent-to-sample ratio of 5:1 (v/w) for 72 hours with intermittent agitation to facilitate extraction [26]. The resulting mixture was filtered through Whatman No. 1 filter paper, and the filtrate was concentrated under reduced pressure using a rotary evaporator. The concentrated extract was subsequently freeze-dried to obtain ITREE. The dried extract was stored in airtight containers under refrigeration until required for the experiment [27].

2.5. Experimental design and extract administration: The freeze-dried extract was reconstituted in distilled water immediately before administration. The animals were randomly allocated into four experimental groups (n = 3 per group). Group A served as the control and received distilled water, whereas Groups B, C, and D received ITREE orally by gavage at doses of 200, 500, and 1000 mg/kg body weight, respectively, once daily for 28 consecutive days. The treatment regimen is presented in Table 1.

Table 1: Dosage of extract administration

Groups	Dosage of ITREE
A (Control)	Distilled water
B	200 mg/kg bw
C	500 mg/kg bw
D	1000 mg/kg bw

2.6. Blood collection: At the end of the 28-day treatment period, the animals were humanely euthanized by cervical dislocation. Whole blood samples were collected by cardiac puncture into appropriately labelled ethylenediaminetetraacetic acid (EDTA)-coated tubes for haematological analysis [28, 29].

2.7. Haematological analysis: Complete blood count (CBC) and erythrocyte, leukocyte, and platelet indices were determined using an automated hematology analyzer in accordance with the manufacturer's operating instructions.

2.8. Statistical analysis: Data are presented as mean ± standard error of the mean (SEM). Statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS) software, version 25.0 (IBM Corp., Armonk, NY, USA). Comparisons among experimental groups were carried out using one-way analysis of variance (ANOVA), followed by the Least Significant Difference (LSD) post-hoc test for multiple comparisons. A *p*-value of less than 0.05 was considered statistically significant.

4. DISCUSSION

Blood is a dynamic, continuously circulating connective tissue that plays essential roles in the transport of oxygen, nutrients, hormones, and metabolic waste products throughout the body [30]. Consequently, haematological parameters are widely regarded as sensitive

indicators in toxicological investigations because they reflect the functional integrity of the hematopoietic system and provide valuable information on physiological and pathological responses to xenobiotics. Parameters such as red blood cell (RBC) count, haemoglobin (Hb) concentration, haematocrit (HCT), white blood cell (WBC) count, mean corpuscular volume (MCV), and mean corpuscular haemoglobin concentration (MCHC) are tightly regulated under normal physiological conditions. Alterations in these indices may therefore indicate disturbances in haematopoiesis, immune function, metabolic homeostasis, or systemic toxicity [25]. Accordingly, the present study evaluated the haematological effects of sub-acute oral administration of ITREE in Wistar rats following a 28-day exposure period.

Table 2: Effects of ITREE on leukocyte and its differentials

Parameter	Group I (Control)	Group II	Group III	Group IV
WBC (×10⁹/L)	4.00 ± 1.20	4.40 ± 0.05	4.00 ± 0.20	5.40 ± 0.08
LYM# (×10⁹/L)	3.57 ± 1.02	4.13 ± 1.22	4.27 ± 1.51	7.10 ± 2.96*
GRAN# (×10⁹/L)	0.23 ± 0.08	0.14 ± 0.05	0.39 ± 0.21	0.24 ± 0.08
MID# (×10⁹/L)	0.20 ± 0.10	0.20 ± 0.06	0.17 ± 0.07	0.27 ± 0.12

*statistically significant difference with Control. Data were expressed as mean ± SEM.

Table 3: Effects of ITREE on erythrocyte indices

Parameter	Group I (Control)	Group II	Group III	Group IV
RBC (x 10^{12}/L)	7.59 ± 0.24	7.34 ± 0.17	7.63 ± 0.10	7.73 ± 0.21
HGB (g/dL)	16.00 ± 0.26	15.33 ± 0.55	15.93 ± 0.26	16.13 ± 0.58
HCT (%)	47.33 ± 1.74	44.23 ± 2.18*	43.30 ± 1.13*	43.43 ± 1.65*
MCV (fl)	62.36 ± 1.08	59.97 ± 1.29*	56.83 ± 1.12*	56.27 ± 0.83*
MCH (pg/Cell)	21.00 ± 0.45	20.77 ± 0.58	20.87 ± 0.38	20.83 ± 0.48
MCHC (g/dL)	33.83 ± 0.73	34.70 ± 1.05	36.83 ± 1.40*	37.08 ± 0.56*
RDW-SD (fl)	49.16 ± 1.24	46.30 ± 0.70	43.43 ± 0.73	36.47 ± 9.50*
RDW-CV (%)	18.20 ± 0.50	17.73 ± 0.55	17.37 ± 0.24	18.10 ± 0.35

*statistically significant difference with Control. Data were expressed as mean ± SEM.

Table 4: Effects of FEREE on platelet indices

Parameter	Group I (Control)	Group II	Group III	Group IV
PLT (×10⁹/L)	832.33 ± 94.33	981.00 ± 321.42	891.00 ± 220.64	998.00 ± 102.10
PCT (%)	0.85 ± 0.09	1.05 ± 0.39	1.19 ± 0.42	1.08 ± 0.14
MPV (fl)	10.30 ± 0.06	10.27 ± 0.79	10.67 ± 0.41	10.77 ± 0.35
PDW	14.00 ± 0.83	14.63 ± 0.77	14.50 ± 1.10	14.03 ± 0.69
P-LCR (%)	24.96 ± 0.47	24.93 ± 4.77	27.67 ± 2.66	28.93 ± 2.55

*statistically significant difference with Control. Data were expressed as mean ± SEM.

The results demonstrated that total WBC count remained comparable with the control group across all treatment doses, indicating that ITREE did not elicit a marked inflammatory or immunotoxic response under the conditions of this study. Nevertheless, a significant increase in lymphocyte count was observed in animals administered the highest dose (1000 mg/kg). Although the mechanism underlying this response was not investigated, it may reflect a mild immunomodulatory effect or enhanced lymphocyte proliferation. Similar increases in lymphocyte counts have been reported following administration of several medicinal plant extracts possessing immunomodulatory phytochemicals, suggesting that certain

bioactive constituents may influence immune cell proliferation and function [31–34]. However, further mechanistic studies are required to substantiate this observation.

Evaluation of erythrocyte parameters showed that RBC count and haemoglobin concentration were not significantly affected by ITREE administration, suggesting that the oxygen-carrying capacity of circulating erythrocytes remained largely preserved throughout the study period. In contrast, significant reductions in haematocrit and mean corpuscular volume were observed in animals receiving the medium and high doses of the extract, accompanied by a significant increase in MCHC at the highest dose. Collectively, these findings suggest dose-dependent alterations in erythrocyte indices that may be consistent with early microcytic changes in red cell morphology, although overt anaemia was not evident.

The reductions in MCV and haematocrit observed at higher doses may indicate alterations in erythrocyte production or maturation. Previous phytochemical studies have shown that *I. trichantha* contains tannins, alkaloids, and other bioactive compounds [36 – 39], some of which have been associated with altered iron utilization, erythropoiesis, or oxidative stress in experimental models [40 – 43]. Nevertheless, because the present study did not quantify these phytochemicals or assess iron metabolism, oxidative stress, or bone marrow function, any mechanistic explanation should be interpreted with caution. Future investigations incorporating biochemical, molecular, and histopathological analyses are therefore required to elucidate the mechanisms responsible for the observed haematological alterations.

The dose-dependent reduction in red cell distribution width-standard deviation (RDW-SD) further suggests a tendency towards a more homogeneous erythrocyte population, which is consistent with the observed reduction in MCV. Platelet indices, including platelet count (PLT), plateletcrit (PCT), mean platelet volume (MPV), platelet distribution width (PDW), and platelet-large cell ratio (P-LCR), exhibited slight increases in the treated groups; however, these differences were not statistically significant. These findings indicate that ITREE did not adversely affect platelet production or platelet morphology during the period of exposure.

Overall, the haematological findings suggest that sub-acute administration of ITREE does not induce marked systemic haematological toxicity at lower doses. However, the dose-dependent alterations observed in selected erythrocyte indices at higher doses indicate that prolonged exposure or administration at elevated doses may influence erythrocyte morphology and hematopoietic function. These observations underscore the importance of dose consideration in the potential therapeutic use of *I. trichantha* and warrant further investigations to establish its long-term haematological safety.

5. CONCLUSION

The present study demonstrated that sub-acute oral administration of ITREE at 200 mg/kg did not produce significant alterations in major haematological parameters, indicating preservation of normal erythropoietic and immune functions under the experimental conditions. In contrast, administration of higher doses (500 and 1000 mg/kg) resulted in significant reductions in haematocrit and mean corpuscular volume, together with an increase in mean corpuscular haemoglobin concentration, suggesting dose-dependent alterations in erythrocyte indices that may represent early microcytic changes.

Although these findings do not indicate severe haematological toxicity following short-term exposure, they highlight the potential for subtle dose-related effects on erythrocyte morphology at higher levels of administration. Consequently, the use of ITREE at elevated doses should be approached with caution until its long-term safety has been fully established. Further studies involving chronic toxicity evaluation, iron metabolism, oxidative stress biomarkers, inflammatory mediators, and

histopathological examination of hematopoietic organs, including the bone marrow and spleen, are recommended to clarify the mechanisms underlying these haematological changes and to provide a more comprehensive assessment of the safety profile of *I. trichantha.*

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